Mohd Shami B.Tech-2027

AI/ML Engineer | Python | Deep Learning | NLP | Computer Vision | Scikit-learn | XGBoost | Flask | Model Deployment

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of experience building **end-to-end AI and machine learning systems** spanning **supervised learning, NLP, computer vision, deep learning (CNN), feature engineering, model optimisation, and Flask-based production deployment**. Proficient in **Python, Scikit-learn, XGBoost, Random Forest, CNN, TF-IDF, SMOTE, SQL, Docker, and Git**. Delivered AI-powered applications achieving up to **97.8% NLP accuracy**, improved clinical diagnostic models by **13 percentage points (AUC-ROC: 0.91)**, and deployed **real-time ML inference APIs**. Seeking an **AI/ML Engineer, AI Engineer Fresher,** or **ML Engineer** role to build and scale intelligent systems.

TECHNICAL SKILLS

AI/ML Algorithms	XGBoost, Random Forest, Logistic Regression, Decision Trees, KNN, Naive Bayes, SVM (basic), Ensemble Methods
Deep Learning	CNN (image classification, feature extraction), TF-IDF (NLP text representation), Keras (basic), Neural Networks
NLP	Text Preprocessing, TF-IDF Vectorisation, Logistic Regression Classifier, SMOTE, Tokenisation, Stop-word Removal
ML Engineering	Scikit-learn Pipelines, Feature Engineering, Data Preprocessing, SMOTE, Hyperparameter Tuning, Cross-Validation
Model Evaluation	AUC-ROC, F1-Score, Precision, Recall, Confusion Matrix, Learning Curves, Bias-Variance Trade-off, Grid Search
Deployment & Stack	Flask (REST API), Docker, Git, Python (Pandas, NumPy), SQL (MySQL), Linux, Jupyter Notebook, ETL Pipelines

WORK EXPERIENCE

Data Science Intern — Codec Technologies India *Aug 2025 – Oct 2025*

- Engineered an **XGBoost AI diagnostic model** on a clinical dataset (5,000+ records); applied **advanced feature engineering, SMOTE class balancing, and Bayesian hyperparameter tuning** to improve classification accuracy from **72% to 85%** and reduce the false-negative rate by **18%**.
- Built end-to-end **Scikit-learn ML Pipelines** integrating data preprocessing, feature scaling, SMOTE oversampling, model training, and cross-validated evaluation into a single reproducible, production-grade workflow.
- Evaluated AI model performance using **AUC-ROC (0.89), F1-Score, Precision/Recall curves, and Confusion Matrix**; delivered structured stakeholder reports with model KPIs, generalisation metrics, and improvement recommendations.
- Implemented **Git-versioned, modular Python training and inference scripts**, enabling reproducible AI experiments and seamless team collaboration across the full ML development lifecycle.

Python Developer Intern — CodTech IT Solutions Pvt. Ltd. *Jun 2025 – Aug 2025*

- Developed **Python ETL pipelines** (Pandas, NumPy) to ingest, clean, and transform data from 4+ sources into model-ready datasets, reducing AI pipeline preprocessing time by **50%**.
- Automated **data validation and quality assurance checks** for ML-ready training datasets, ensuring feature consistency, schema integrity, and zero data leakage across train/validation/test splits.

PROJECTS

Revive — AI-Powered Clinical Disease Prediction System *Jan 2026 – Mar 2026*

Python, XGBoost, Scikit-learn Pipelines, SMOTE, Flask REST API, Docker — github.com/mohdshamii/Revive

- Built a **production AI system** on UCI Heart Disease dataset (303 records): end-to-end ML pipeline with feature engineering, SMOTE balancing, XGBoost training, and k-fold cross-validation achieving **85% accuracy (AUC-ROC: 0.91, F1: 0.87)**.
- Deployed the AI model as a **Flask REST API** with real-time JSON inference endpoints and probability scores; containerised with **Docker** for reproducible, environment-agnostic production deployment.

HydroRaksh (JalSuraksha) — AI-Driven Groundwater Intelligence Platform *Jul 2026 – Oct 2026*

Python, Flask, FastAPI, Scikit-learn, SHAP, SQLAlchemy, Docker, Jupyter — github.com/mohdshamii/HydroRaksh

- Built a **pan-India groundwater decision-support platform** (36 States/UTs, 700+ districts) ingesting public CGWB, IMD, India-WRIS and CWC data, with **16 reproducible Jupyter notebooks** covering EDA, validation, feature engineering, forecasting and **SHAP explainability**.
- Trained water-table forecasters (Ridge, Random Forest, Gradient Boosting) with **rolling-origin cross-validation**, cutting MAE from **29.85 m** (naive baseline) to **3.85 m**; built a hydro-geological **collaborative-filtering recommender** for recharge interventions (NDCG@5: 0.437).
- Served models through **Flask and FastAPI REST endpoints**, containerised with **Docker**, and backed by **101 automated unit tests**.

FarmAIQ — Multi-Model AI Platform for Smart Agriculture *Nov 2025 – Dec 2025*

Python, Random Forest, CNN, Deep Learning, Cross-Validation, Grid Search — github.com/mohdshamii/FarmAIQ

- Designed a **dual-model AI platform**: a **Random Forest crop recommendation engine** (Kaggle dataset, 2,200 records, **93% accuracy**) and a **CNN-based plant disease classifier** (PlantVillage dataset) using deep learning feature extraction; applied grid-search tuning and cross-validation to optimise generalisation on unseen agricultural data.

EDUCATION

B.Tech in Data Science — Teerthanker Mahaveer University, Moradabad, UP *Expected 2027*

CGPA: 8.5 / 10

CERTIFICATIONS

Simplilearn: Deep Learning & ML with Python (2025) | **IBM:** Data Analysis & Problem Solving (2026) | **ISRO:** Scientific Observation from Space (2026)

HackerRank: Python, SQL & Problem Solving (2025) | **IIT Bombay:** Intro to Computer Science (2024) | **ITM EdTech:** Advanced Data Analysis – Excel & Power BI (2025)

ACHIEVEMENTS & LANGUAGES

Smart India Hackathon (SIH): Selected for college round | **India AI Impact Summit:** Presented AI/ML project to industry leaders | **850+ DSA problems** solved in Python (HackerRank & LeetCode)

Languages: Hindi (Native), English (Professional), Urdu